

Reviewer Pack — Real Root Count Inverse Proportional to SOS Degree for Max-C...

Entry d9f5244dde36 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Real Root Count Inverse Proportional to SOS Degree for Max-CUT

Entry ID: d9f5244dde36 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-09 21:07:54 UTC

1. Conjecture

Field A (mathematical branch): Geometry of Polynomials **Field B** (complexity object): SOS Degree for Max-CUT

Statement:

For a random Max-CUT instance on n variables, the expected number of real roots of the corresponding polynomial $P(x)$ is $\Theta(n^{\{1/d\}})$, where d is the SOS degree required to approximate the maximum cut with a 0.878-approximation.

Rationale (proposer’s reasoning):

The SOS hierarchy’s degree d relates to the precision of the approximation. Polynomials with more real roots may have simpler structures, allowing lower-degree SOS certificates via real-rootedness-based moment matrix rank bounds.

Taxonomy category: SOS_DEGREE (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 49066310ddb6c495

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gaussian_elimination(A, b):
    n = len(b)
    for i in range(n):
        max_row = max(range(i, n), key=lambda r: abs(A[r][i]))
        A[i], A[max_row] = A[max_row], A[i]
        b[i], b[max_row] = b[max_row], b[i]
        lead_coeff = A[i][i]
        for j in range(n):
            A[i][j] /= lead_coeff
        b[i] /= lead_coeff
        for k in range(n):
            if k != i:
                factor = A[k][i]
                for j in range(n):
                    A[k][j] -= factor * A[i][j]
                b[k] -= factor * b[i]
    return A, b

def matrix_multiply(A, B):
    n = len(A)
    C = [[0] * n for _ in range(n)]
    for i in range(n):
        for j in range(n):
            for k in range(n):
                C[i][j] += A[i][k] * B[k][j]
    return C

def determinant(A):
    if len(A) == 1:
        return A[0][0]
    det = 0
    sign = 1
    for j in range(len(A)):
        submatrix = [row[:j] + row[j+1:] for row in A[1:]]
        det += sign * A[0][j] * determinant(submatrix)
        sign *= -1
```

```

    return det

def sturm_sequence(coefficients):
    n = len(coefficients)
    if n < 2:
        raise ValueError("Sturm sequence requires at least two coefficients")

    seq = [coefficients[:]]
    while True:
        next_coeff = [-seq[-2][i+1] * (len(seq[-1]) - i - 1) / seq[-1][0] for i in range(len(seq[-1]) - 1)]
        if all(coeff == 0 for coeff in next_coeff):
            break
        seq.append(next_coeff)

    return seq

def count_real_roots(sturm_seq):
    n = len(sturm_seq)
    sign_changes = [sum(1 for i in range(n-1) if sturm_seq[i][j] * sturm_seq[i+1][j] < 0) for j in range(n)]
    return sum(sign_changes)

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = random.randint(5, 40)
    edge_weights = [[random.uniform(0.1, 1.0) for _ in range(n)] for _ in range(n)]
    variables = [random.choice([-1, 1]) for _ in range(n)]

    polynomial_coeffs = []
    for i in range(n):
        for j in range(i+1, n):
            coeff = edge_weights[i][j]
            if coeff > 0:
                polynomial_coeffs.append(coeff * (variables[i] + variables[j]))

    sturm_seq = sturm_sequence(polynomial_coeffs)
    real_roots_count = count_real_roots(sturm_seq)

    return {
        "metric_name": "real_root_count",
        "metric_value": real_roots_count,
        "instances_tested": 1,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 3**i for i in range(5, 8)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

```

```

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
elif any(not r["conjecture_holds"] for r in results):
    first_failing_seed = next(seed for seed, r in zip(seeds, results) if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")
else:
    print(f"RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c564256f.py", line 106, in <module>
    result = run_trial(seed)
              ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c564256f.py", line 89, in run_trial
    sturm_seq = sturm_sequence(polynomial_coeffs)
                  ~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c564256f.py", line 63, in sturm_sequence
    next_coeff = [-seq[-2][i+1] * (len(seq[-1]) - i - 1) / seq[-1][0] for i in range(len(seq[-1]) - 1)]
                                                         ~~~~~
IndexError: list index out of range

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed during Sturm sequence computation, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the IndexError in sturm_sequence() by adding input validation for polynomial coefficients

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	91616
2	propose	ollama_remote	qwen3:8b	0	0	84886
3	preregistration	ollama_remote	qwen3:8b	0	0	24322
4	novelty	ollama_remote	qwen3:8b	0	0	21423
5	novelty	ollama_remote	qwen3:8b	0	0	13887
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11629
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11743
8	judge	ollama_remote	qwen3:8b	0	0	17011

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 276517 ms total latency. Provider mix: {'ollama_remote': 8}

(full prompt+response transcripts available in `research/audit/d9f5244dde36.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/d9f5244dde36.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/d9f5244dde36.tar.gz` (if generated)